Brucella canis Brucellosis (Canine Brucellosis): A Comprehensive Disease Characterization

Disease: Brucella canis Brucellosis **MONDO ID:** MONDO:0001857 **ICD-10:** A23.3 (Brucellosis due to Brucella canis) · **ICD-11:** 1B95 · **MeSH:** Brucellosis, Canine **Pathogen:** *Brucella canis* (NCBI:txid36855) **Category:** Infectious Disease (bacterial zoonosis)

Summary

Brucella canis brucellosis is a chronic, zoonotic bacterial infection whose natural host is the domestic dog (*Canis lupus familiaris*, NCBI:txid9615). The causative agent, *Brucella canis*, is a naturally-rough (rough-LPS), facultatively intracellular Gram-negative α-proteobacterium that is phylogenetically nested within *Brucella suis* despite its distinct clinical niche. In dogs the infection presents primarily as reproductive failure — late-gestation abortion, epididymitis, prostatitis, and infertility — but it also causes discospondylitis (vertebral disc/endplate infection), anterior uveitis/endophthalmitis, and frequently subclinical, persistent bacteremia. Transmission is predominantly venereal and through reproductive fluids and aborted material, placing breeding kennels and free-roaming stray populations at highest risk. The organism is an under-recognized zoonosis: humans acquire infection through contact with infected dogs, and standard smooth-*Brucella* human serology fails to detect it.

Mechanistically, the disease is driven by the pathogen's capacity to survive and replicate inside host macrophages. *B. canis* enters cells through cholesterol- and ganglioside-GM1-rich lipid rafts, elicits only weak macrophage activation, and establishes a *Brucella*-containing vacuole (BCV) that traffics to the endoplasmic reticulum — a process orchestrated by lipopolysaccharide (LPS), the VirB Type IV secretion system (T4SS), and the BvrR/BvrS two-component regulatory system. This intracellular, immune-evasive lifestyle explains the chronic, relapsing course, the intermittent low-level bacteremia that undermines diagnosis, and the notorious difficulty of achieving microbiological cure even with prolonged multidrug antimicrobial therapy.

The public-health and veterinary importance of *B. canis* is rising. There is no vaccine and no reliably curative treatment; control depends on serological screening, test-and-removal, neutering, biosecurity, and — increasingly — import regulation, since the international movement of dogs is spreading the organism into previously low-prevalence regions. This report synthesizes 12 confirmed findings across 28 reviewed papers into a section-by-section knowledge-base entry, with ontology mappings and cited primary evidence. Because this is an infectious rather than heritable disease, the genetic/variant/inheritance portions of the template are largely **Not applicable** at the host level and are annotated as such; the corresponding "molecular genetics" reside in the *pathogen* genome.

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1. Disease Information

Overview. *Brucella canis* brucellosis is a contagious bacterial disease of dogs and an occupational/companion-animal zoonosis of humans. In dogs it is chiefly a reproductive disease and a cause of discospondylitis, with a substantial proportion of subclinical infections. As one authoritative case report summarizes, "*Brucella canis* is a zoonotic bacterium, causing mainly reproductive disorders and discospondylitis in dogs, but infections may also be subclinical. Mating is one of the main transmission routes, putting breeding dogs and stray dogs at risk of infection" (PMID: 41957789).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0001857
ICD-10	A23.3 (Brucellosis due to *Brucella canis*)
ICD-11	1B95 (Brucellosis)
MeSH	Brucellosis, Canine
Pathogen taxon	NCBI:txid36855 (*Brucella canis*)
OMIM / Orphanet	Not applicable (infectious, non-Mendelian disease)

Synonyms / alternative names. Canine brucellosis; *B. canis* infection; contagious canine abortion (historical, in kennels). The type strain is **RM6/66**.

Nature of the information. All evidence is derived from **aggregated disease-level resources** — veterinary case series, kennel outbreak investigations, seroprevalence surveys, and human seroepidemiological studies — rather than individual EHR-level patient records. Evidence source types span veterinary clinical, human clinical/seroepidemiological, in vitro (macrophage infection models), and computational (whole-genome phylogenetics).

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2. Etiology

Primary cause — infectious. The disease is caused entirely by infection with *Brucella canis*, a Gram-negative coccobacillus. It is not a genetic disease; there are no causal host genes, no pathogenic germline/somatic variants, and no Mendelian inheritance. Accordingly, Section 4 (host genetic/molecular information) and much of Section 9's inheritance sub-items are **not applicable** — the "genetic" story of this disease is that of the *pathogen's* genome (Section 7-genomics and Section 6).

Transmission routes (principal risk exposures). Mating (venereal) is a primary route; other routes include contact with aborted materials and vaginal discharge, infected semen, oronasal contact with contaminated fluids, and vertical (transplacental/milk) transmission. "Mating is one of the main transmission routes, putting breeding dogs and stray dogs at risk of infection" (PMID: 41957789).

Risk factors (host/environmental). - **Reproductive status:** intact (unaltered) dogs are at higher risk than neutered/spayed dogs. - **Age:** older dogs show higher seropositivity in stray populations. - **Living conditions:** stray, shelter, and breeding-kennel dogs; peridomestic/free-roaming populations in socioeconomically deprived urban areas. - **Occupational (human):** kennel employees, veterinarians, laboratory diagnosticians, animal-care providers. - **Importation:** movement of dogs from endemic regions.

From a South Dakota stray-dog survey: "Older dogs were significantly more likely to be *B. canis*-positive than younger dogs, as were reproductively intact dogs versus altered dogs" (PMID: 33002657).

Protective factors. Neutering/spaying reduces risk (venereal exposure reduction). No genetic protective variants are described (not applicable). There is no vaccine (Section 13).

Gene–environment interactions. Not applicable at the host level. The relevant "gene–environment" axis is the pathogen's virulence-gene repertoire interacting with the host intracellular environment (Section 6).

3. Phenotypes

B. canis brucellosis produces distinct clinical syndromes in dogs. Because this is a non-Mendelian infectious disease, HPO terms (human-phenotype-oriented) are suggested by analogy where a human-analogous phenotype exists.

Phenotype	Type	Onset / course	Frequency	Suggested HPO
Late-gestation abortion (≈ day 45–59)	Clinical sign (reproductive)	Adult breeding animals; episodic	Common in bitches	HP:0100603 (Stillbirth, analogous)
Epididymitis / orchitis / prostatitis; infertility	Clinical sign (reproductive)	Adult intact males; chronic	Common in infected males	HP:0000029 (Abnormality of the testis)
Discospondylitis (back pain, lameness, spinal pain)	Clinical sign (musculoskeletal)	Young intact dogs (median ~2.6 yr); chronic/ progressive	Recognized cause	HP:0003418 (Back pain)
Anterior uveitis / endophthalmitis (unilateral)	Clinical sign (ocular)	Chronic/ recurrent	Uncommon	HP:0000554 (Uveitis)
Lymphadenopathy	Clinical sign	Variable	Common	HP:0002716 (Lymphadenopathy)
Subclinical persistent bacteremia	Laboratory/ microbiological	Insidious; chronic	Frequent	—
Fever / malaise (mainly human)	Symptom	Nonspecific, insidious	Common in symptomatic humans	HP:0001945 (Fever), HP:0025406 (Malaise)

Characteristics. Onset is typically adult (reproductive disease) or young-adult (discospondylitis); course is chronic, insidious, and relapsing; severity ranges from subclinical to severe (endocarditis in rare human cases). The reproductive phenotype is episodic (tied to gestation/breeding). *B. canis* discospondylitis is distinguished clinically by younger age and characteristic spinal distribution (Section 6/clinical).

Quality-of-life impact. In dogs: reproductive failure ends breeding value; discospondylitis causes chronic pain and lameness; uveitis threatens vision. In humans: chronic febrile illness with fatigue; rare severe complications. Formal QoL instruments (EQ-5D/SF-36) have not been applied to this zoonosis.

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4. Genetic/Molecular Information (Host)

Not applicable. *B. canis* brucellosis is an acquired infectious disease with no causal host genes, pathogenic variants, modifier genes, host epigenetic drivers, or chromosomal abnormalities. Host genetic susceptibility loci have not been established. The molecular "genetics" of the disease reside in the pathogen genome.

Pathogen genomics (analogous content). Whole-genome SNP phylogeny of the highly conserved *Brucella* genus (20,154 shared orthologous SNPs) shows that "*B. suis* was determined to be paraphyletic in our analyses, only forming a monophyletic clade when the *B. canis* genome was included" (PMID: 19201792). Core-genome MLST places *B. canis* "closest relationship with *B. suis* bv. 4, followed by *B. suis* bv. 3 and bv. 1" (PMID: 30385393). Divergence dating indicates "most *Brucella* species diverged from their common *B. ovis* ancestor in the past 86,000 to 296,000 years, which precedes the domestication of their livestock hosts" (PMID: 19201792).

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5. Environmental Information

- **Environmental factors:** Contaminated kennel environments, aborted materials, vaginal/seminal discharges, and shared breeding facilities are the principal fomite/exposure reservoirs. There is no toxin, radiation, or pollution etiology.

- **Lifestyle factors (human):** Occupational animal contact (kennel work, veterinary practice, laboratory handling); keeping infected dogs in the home. "Humans can become infected with *B canis* when an infected pet dog is brought into their home" (PMID: 36892988).
- **Infectious agent:** *Brucella canis* (NCBI:txid36855), a naturally-rough member of the *B. suis* clade. Type strain RM6/66. Key virulence molecule: lipopolysaccharide (LPS, CHEBI:16412).

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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Exposure** to infected reproductive fluids/aborted material or venereal contact **leads to** mucosal entry of *B. canis*.
2. Bacteria are taken up by macrophages and dendritic cells; entry **occurs via** cholesterol- and ganglioside-GM1-rich **lipid rafts** and the class A scavenger receptor **→ results in** intracellular localization (demonstrated in murine macrophages for *B. canis* and *B. ovis*).
3. The organism **induces only weak macrophage activation** (rough *Brucella* LPS is poorly stimulatory) **→ results in** evasion of early bactericidal responses.
4. Coordinated action of **LPS, the VirB Type IV secretion system (T4SS), and the BvrR/BvrS two-component system leads to** formation of an early and then late **Brucella-containing vacuole (BCV)** that avoids lysosomal fusion.
5. The BCV **traffics to and interacts with the endoplasmic reticulum (ER) → results in** a replication-permissive niche where bacteria multiply (upstream driver of persistence).
6. Intracellular replication with slow release **leads to** chronic, **intermittent low-level bacteremia** (downstream consequence; also the reason diagnosis is hard).
7. Bacteremia **seeds** target tissues — **reproductive tract** (placenta/trophoblast, epididymis, prostate), **intervertebral discs/vertebral endplates**, and the **eye (uvea)** — **→ results in** the clinical syndromes (branch point: reproductive vs. skeletal vs. ocular disease).
8. In the placenta, trophoblast infection **leads to necrotic placentitis → results in** late-gestation abortion (demonstrated by immunohistochemical trophoblast staining).
9. In humans, the same intracellular-persistence mechanism **leads to** nonspecific febrile illness and, rarely, **infective endocarditis** and neurological complications (inferred to share the canine intracellular mechanism).

Detail by category

Molecular pathways / virulence machinery. "Major virulence factors are: lipopolysaccharide (LPS), T4SS secretion system and BvrR/BvrS system, which allow interaction with host cell surface, formation of an early, late BCV (Brucella Containing Vacuole) and interaction with endoplasmic reticulum (ER) when the bacteria multiply" (PMID: 30015453). The overarching pathogenic property: "The most important feature of *Brucella* is the ability to survive and multiply within both phagocytic and non-phagocytic cells" (PMID: 30015453).

Cellular processes / entry. "The establishment of infection by *Brucella ovis* and *Brucella canis* in J774.A1 macrophages was found to be dependent upon cholesterol and ganglioside GM(1), two components of lipid rafts" (PMID: 20083220), and "both bacteria induced a weak degree of macrophage activation" (PMID: 20083220) — i.e., immune-evasive entry favoring persistence.

Immune system involvement. Chronic intracellular infection with dampened innate activation; delayed/insufficient sterilizing immunity underlies relapse. Rough LPS is a weaker TLR4 agonist than smooth LPS, contributing to stealth.

Tissue damage mechanisms. Granulomatous/necrotizing inflammation at seeded sites; placental necrosis with trophoblast tropism drives abortion; discitis/osteomyelitis at vertebral endplates; chronic uveal inflammation.

Suggested ontology terms. - GO biological process: GO:0051701 (biological process involved in interaction with host); GO:0044409 (entry into host); intracellular replication in ER-derived vacuole. - GO cellular component: GO:0005783 (endoplasmic reticulum) — BCV target; phagosome/endosome-derived vacuole (GO:0005768). - CL cell types: CL:0000235 (macrophage), CL:0000451 (dendritic cell), trophoblast (placental target). - CHEBI: LPS (CHEBI:16412), cholesterol (CHEBI:16113), ganglioside GM1 (CHEBI:18216).

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7. Anatomical Structures Affected

Level	Structures	Suggested ontology
Primary organs	Male reproductive tract (testis, epididymis, prostate); female reproductive tract (uterus, placenta); intervertebral discs/vertebrae; eye (uvea)	UBERON:0000473 (testis), UBERON:0001301 (epididymis), UBERON:0002367 (prostate), UBERON:0001987 (placenta), UBERON:0002412 (vertebra), UBERON:0001803 (uvea)
Secondary/ systemic	Lymph nodes (lymphadenopathy), spleen; heart valves (human endocarditis)	UBERON:0000029 (lymph node), UBERON:0002106 (spleen), UBERON:0006618 (heart valve)
Body systems	Reproductive, musculoskeletal, ocular, lymphoreticular; cardiovascular (rare human)	—
Tissue/cell	Placental trophoblast; macrophages/ dendritic cells (reservoir)	CL:0000235, CL:0000451
Subcellular	Endoplasmic reticulum (replication niche); Brucella-containing vacuole	GO:0005783
Localization	Discospondylitis often focal; uveitis characteristically **unilateral**; C2–C5 cervical involvement over-represented	—

Placental trophoblast tropism is directly documented: "necrotic placentitis and the strong immunostaining of trophoblast cells by immunohistochemistry" (PMID: 21217047). Cervical predilection in discospondylitis: "higher odds of C2-C5 involvement (OR = 13.0; 95% CI 1.6-106.7)" (PMID: 40317976).

8. Temporal Development

- **Onset.** Adult onset for reproductive disease; young-adult onset for discospondylitis (median 2.6 yr). Pattern is **insidious/chronic**; many infections subclinical for extended periods.

- **Progression.** Chronic, **relapsing/episodic** (reproductive events; recurrent uveitis). Persistent intracellular infection with intermittent bacteremia can last months to years. Discospondylitis is progressive if untreated.
- **Duration.** Effectively **lifelong** without successful eradication; microbiological cure is difficult and relapse common.
- **Remission/critical windows.** Antimicrobial therapy can induce clinical remission and declining serology, but relapse is frequent after cessation; neutering reduces shedding. A ≥40% decline in *B. canis* antibody value at 2–6 months supports treatment response: "A 40% decline in CBM assay values 2 to 6 months posttreatment can be supportive of response to treatment" (PMID: 36892988). In ocular disease, seronegativity took a "median of 96 weeks (range: 36-112 weeks) of therapy" (PMID: 19392878).

9. Inheritance and Population (Epidemiology)

Inheritance. Not applicable — infectious, non-heritable disease. No penetrance/expressivity/anticipation/founder-effect/consanguinity/carrier-frequency concepts apply to the host.

Epidemiology (seroprevalence).

Population	n	Finding	Source
Stray/surrendered dogs, South Dakota reservations (2015–2019)	3,898	Apparent seroprevalence **6.8%**; adjusted true prevalence **29.4%**; declining over time; older & intact dogs higher risk	PMID: 33002657
Shelter dogs, Mississippi	571	**4.9%** RSAT-seropositive (2.3% confirmed)	PMID: 30314794
Urban slum humans, Salvador, Brazil	174	**4.6%** *B. canis* seropositive	PMID: 23000868
Dog-exposed adults (occupational), USA	306 vs 101 controls	**10.8% (RSAT) – 3.6% (ME-RSAT)**; kennel workers OR 2.7 (95% CI 1.3–5.8)	PMID: 24751191

Direct quotes: "we determined an overall apparent *B. canis* seroprevalence of 6.8% (adjusted estimated true prevalence of 29.4%)... Older dogs were significantly more likely to be *B. canis*-positive than younger dogs, as were reproductively intact dogs versus altered dogs" (PMID: 33002657); "Twenty-eight dogs (4.9%) were seropositive for *B. canis* antibodies on the RSAT" (PMID: 30314794).

Geographic distribution. Endemic in the Americas (notably parts of the US South, Latin America), and increasingly reported in Europe via dog importation. Whole-genome/MLVA typing has traced international spread: "We detected three recent introductions of different *B. canis* strains in kennels: two traced from Mexico and one from Panama" (PMID: 33965789).

Demographics. Highest in intact, older, free-roaming/kennel dogs. Human cases cluster in occupationally exposed individuals and in socioeconomically deprived urban areas with expanding peridomestic dog populations: "Changes in the urban environment due to slum communities' expansion have resulted in increased dog populations in the peridomiciliary environment" (PMID: 23040615).

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10. Diagnostics

Diagnosis is challenging because of the **rough antigen** (standard smooth-*Brucella* serology fails) and **intermittent low-level bacteremia**.

Serology (screening). Rough-antigen assays are required: RSAT, 2ME-RSAT, AGID, and immunochromatographic/CBM tests. Sensitivities are limited:

Test	Sensitivity	Specificity
RSAT	70.6–75%	moderate
2ME-RSAT	31.8–37.5%	up to 100%
AGID	27.8–52.9%	up to 100%
Immunochromatographic (ICT)	89.6%	100%

"The diagnostic sensitivity of RSAT, 2ME-RSAT and AGID was respectively 70.58%, 31.76%, and 52.94%" (PMID: 18656213). The ICT "showed a high diagnostic specificity and a diagnostic sensitivity value greater than that observed in the RSAT, 2ME-RSAT and AGID. However, 10.41% of infected dogs had negative results by ICT" (PMID: 26489371).

Molecular (confirmatory). Blood PCR (IS711 target) markedly outperforms serum PCR: "The sensitivities of blood PCR and serum PCR were, respectively, 97.14 per cent and 25.71 per cent. The specificities of both were 100 per cent" (PMID: 20643887) — establishing **blood** (not serum) as the preferred matrix. Semen PCR is a useful confirmatory test before breeding and is highly sensitive, detecting "as few as 3.8 fg of *B. canis* DNA" (PMID: 17343907).

Culture. Blood/tissue/vaginal culture is confirmatory but insensitive (intermittent bacteremia) and is an **occupational hazard requiring BSL-3** handling. In one outbreak, culture succeeded in "only 3 of 31 (10%) cases" (PMID: 21217047).

Histopathology/IHC. Placental IHC (anti-*B. canis*) demonstrates trophoblast infection and is a rapid, sensitive adjunct in abortion cases (PMID: 21217047).

Imaging. Radiography/CT/MRI for discospondylitis (osteolysis, endplate sclerosis, periosteal proliferation); ophthalmic exam for uveitis/endophthalmitis.

Human diagnosis caveat. "Human serological assays designed to detect brucellosis will miss infections caused by *Brucella canis*, and low levels of periodic bacteremia limit diagnosis by blood culture" (PMID: 24751191).

Differential diagnosis. Discospondylitis from other agents (*Staphylococcus*, *Streptococcus*, *E. coli*, fungal); other causes of canine infertility/abortion; other *Brucella* spp. in humans.

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11. Outcome / Prognosis

- **Mortality.** Low direct mortality in dogs; the disease is chronic rather than rapidly fatal. In humans, mortality is low but severe complications (infective endocarditis, neurological) occur.
- **Morbidity.** Substantial: permanent loss of reproductive function, chronic spinal pain/lameness, potential vision loss. Infected breeding animals are typically removed from breeding.
- **Recovery / cure.** Microbiological cure is difficult; relapse is common even after prolonged multidrug therapy. Serologic monitoring (≥40% antibody decline at 2–6 months) supports response (PMID: 36892988); ocular cases required a median 96 weeks to seronegativity (PMID: 19392878).
- **Prognostic factors.** Site of disease, chronicity, treatment adherence/duration, and neutering status. Declining rough-antigen antibody titer is a favorable prognostic marker.

- **Human severe outcomes.** "We report *Brucella canis* human infection link to Gaucher's disease and Guillain Barré syndrome and discuss complications observed in a case with infective endocarditis" (PMID: 23040615).

12. Treatment

Pharmacotherapy — combination antimicrobials. Because *B. canis* is intracellular, **poly-antimicrobial, prolonged therapy** is standard; monotherapy fails. In a series of 30 naturally infected dogs, "poly-antimicrobial therapy was prescribed in 97% (29/30) of cases" (PMID: 36892988). Common agents (suggested NCIT clinical-intervention terms in parentheses):

Drug	Class	NCIT (suggested)
Doxycycline / tetracyclines	Tetracycline antibiotic	NCIT:C299 (Doxycycline)
Streptomycin / gentamicin	Aminoglycoside	NCIT:C864 (Streptomycin)
Rifampin	Rifamycin	NCIT:C793 (Rifampin)
Enrofloxacin	Fluoroquinolone (veterinary)	—

A validated ocular regimen combined four drugs: "a novel antimicrobial protocol that included doxycycline, enrofloxacin, rifampin and streptomycin" (PMID: 19392878), with "serological values declined with seronegativity achieved in all dogs after a median of 96 weeks (range: 36-112 weeks) of therapy" (PMID: 19392878).

Adjunct/interventional. Neutering (surgical) reduces shedding and venereal transmission and is a standard component of management (NCIT:C15329 Castration / ovariohysterectomy). **No gene, cell, RNA, targeted, or immunotherapy** applies (not applicable).

Treatment outcomes. High relapse; combination therapy with prolonged duration and serologic monitoring is required. Because "*Brucella* spp… evade antimicrobial therapy," complete eradication is uncertain, and euthanasia/removal is often chosen for breeding-colony control.

Personalized medicine / pharmacogenomics. Not applicable.

13. Prevention

Prevention is the cornerstone because diagnostics and therapeutics are limited: "Diagnosis and treatment of brucellosis is challenging due to the limitations of diagnostic assays and the tendency of *Brucella* spp to produce nonspecific, insidious clinical signs and evade antimicrobial therapy, making prevention essential" (PMID: 36862545).

- **Primary prevention. No vaccine exists.** "*Brucella canis* infection is one of the most important causes of infertility in dogs and is a zoonosis for which no effective treatment or vaccines exist" (PMID: 36155554). Prevention relies on pre-breeding testing, biosecurity, and avoiding exposure to infected animals/materials.
- **Secondary prevention.** Serological screening of breeding stock and pre-mating semen/blood PCR; test-and-removal in kennels; cascade testing of contacts.
- **Tertiary prevention.** Neutering, isolation, treatment with serologic monitoring to reduce shedding and complications.
- **Public health / One Health.** Import control, occupational biosafety (BSL-3 for culture), owner/breeder/veterinarian education. Awareness is variable and control is hampered because it "is a zoonosis for which no effective treatment or vaccines exist" (PMID: 36155554). Import legislation is often inadequate: "Current legislation is insufficient to address the risks and pre-import control measures are focused only on rabies virus and the fox tapeworm *Echinococcus multilocularis*" (PMID: 41925288).
- **Human exposure routes reported to CDC:** "Human consumption of unpasteurized dairy products and occupational exposure of laboratory diagnosticians, veterinarians, and animal care providers are responsible for human exposures reported to the US CDC" (PMID: 36862545).

14. Other Species / Natural Disease

- **Primary host / taxonomy.** Domestic dog, *Canis lupus familiaris* (NCBI:txid9615). Pathogen: *Brucella canis* (NCBI:txid36855). Wild canids may be susceptible. Serosurveys of wildlife (black bears, PMID: 25075540; free-roaming reserve dogs, PMID: 35687596) were negative, indicating dogs are the principal reservoir rather than wildlife.
- **Breed.** No specific breed predisposition established (VBO not applicable beyond species); risk is driven by breeding/kennel management, not breed genetics.

- **Zoonotic transmission.** Confirmed zoonosis: "Humans can become infected with *B canis* when an infected pet dog is brought into their home" (PMID: 36892988); occupational human infection is documented (PMID: 24751191).
- **Comparative biology.** *B. canis* is one of several *Brucella* spp. of US One Health relevance: species "infect dogs (*B. canis*), swine (*B. suis*), and cattle/bison (*B. abortus*)" (PMID: 36862545). Evolutionarily, *B. canis* is a rough clade nested within *B. suis*, diverged from a common *B. ovis* ancestor (PMID: 19201792).

15. Model Organisms

- **In vitro / cellular models.** Murine macrophage cell lines (J774.A1) are the workhorse for *B. canis* entry/survival studies; these established the lipid-raft (cholesterol, GM1) and scavenger-receptor entry mechanism (PMID: 20083220). Human U937 monocytic cells (RA/VD-differentiated) discriminate virulence: during infection these cells "could discriminate between the human pathogen *Brucella suis* S1, which strongly multiplied, and the animal pathogen *Brucella canis*, which survived without multiplication" (PMID: 8071593) — a basic model of bacteria–human macrophage interaction.
- **Natural/animal model.** The **dog itself** is the definitive natural host and de facto disease model; mouse infection models are used for *Brucella* pathogenesis generally.
- **Phenotype recapitulation / limitations.** Macrophage models capture intracellular survival and entry but not the reproductive, skeletal, and ocular tissue tropism of natural canine disease. No standardized knockout/transgenic host model exists (host genetics not causal).
- **Suggested CL terms:** CL:0000235 (macrophage), CL:0000576 (monocyte).

Mechanistic Model / Interpretation



The unifying theme is **intracellular persistence with immune stealth**. Every downstream clinical and diagnostic feature flows from it: chronicity and relapse (persistence in an ER-derived niche), diagnostic insensitivity (intermittent bacteremia + rough antigen), and therapeutic difficulty (antibiotics must penetrate cells and be sustained for months). Upstream determinants are the lipid-raft entry route and the LPS/T4SS/BvrR-BvrS virulence triad; downstream determinants are tissue tropism to placenta, spine, and eye.

Evidence Base

PMID	Contribution	Evidence type
41957789	Core disease definition, syndromes, transmission	Veterinary clinical (case)
30015453	Intracellular lifestyle; LPS/T4SS/BvrR-BvrS; BCV→ER	Review
20083220	*B. canis*-specific lipid-raft entry; weak activation	In vitro
8071593	U937 macrophage virulence discrimination	In vitro
20643887	Blood PCR 97.1% vs serum 25.7%	Diagnostic
18656213	RSAT/2ME-RSAT/AGID sensitivities	Diagnostic
26489371	ICT performance; 10.4% missed	Diagnostic
17343907	Semen PCR (3.8 fg detection)	Diagnostic
21217047	Necrotic placentitis, trophoblast IHC; kennel outbreak	Veterinary pathology
33002657	Stray-dog seroprevalence, age/intact risk	Epidemiology
30314794	Shelter-dog seroprevalence	Epidemiology
24751191	Human occupational infection; serology misses *B. canis*	Human seroepidemiology
23000868	Human urban-slum seroprevalence 4.6%	Human seroepidemiology
23040615	Human endocarditis, comorbidities, urban risk	Human clinical
36892988	Poly-antimicrobial therapy; serologic monitoring; household zoonosis	Veterinary clinical
40317976	Discospondylitis clinical features (age, C2–C5)	Veterinary case-control
19392878	Ocular disease; 4-drug regimen; 96-wk cure	Veterinary clinical
19201792	Genome phylogeny; *B. canis* within *B. suis*; divergence	Computational genomics
30385393	cgMLST: closest to *B. suis* bv. 4	Computational genomics
33965789	WGS/MLVA outbreak tracing; international spread	Genomic epidemiology

PMID	Contribution	Evidence type
36155554	No vaccine/cure; breeder awareness gap	Survey
41925288	Import-driven emergence; legislative gaps	Review
36862545	US One Health framing; CDC exposure routes; prevention essential	Review

Consistency / conflicts. The evidence base is internally consistent. Wildlife serosurveys (PMID: 25075540, PMID: 35687596) were negative, reinforcing (rather than challenging) that dogs are the reservoir. Diagnostic sensitivity estimates vary across serological tests, consistent with the known limitations of rough-antigen serology.

Limitations and Knowledge Gaps

1. **No host-genetic dimension.** Template sections designed for Mendelian diseases (causal genes, variants, inheritance, penetrance, model knockouts) are not applicable; the "molecular" content is pathogen-centric.
2. **Human epidemiology under-quantified.** True human incidence/prevalence is unknown because standard serology misses *B. canis* and bacteremia is intermittent — likely substantial under-ascertainment.
3. **Treatment evidence is observational.** No randomized trials define the optimal antimicrobial combination or duration; relapse rates and true cure rates are imprecisely known.
4. **No vaccine and no validated correlate of protection.** Immunological mechanisms of protective immunity are poorly defined.
5. **Diagnostic thresholds** (e.g., ≥40% antibody decline) derive from small case series and need prospective validation.
6. **Ontology mapping** for a veterinary zoonosis is imperfect: HPO/CL/UBERON terms are human-oriented and used here by analogy.

Proposed Follow-up Experiments / Actions

1. **Standardize a confirmatory diagnostic algorithm** combining rough-antigen serology (RSAT/ICT) with IS711 **blood** PCR and, pre-breeding, **semen** PCR; prospectively validate the ≥40% antibody-decline cure criterion.
2. **Prospective treatment trial** comparing doxycycline+aminoglycoside vs. four-drug regimens for duration-to-microbiological-cure and relapse, with predefined serologic and PCR endpoints.
3. **Human surveillance study** using rough-antigen–specific assays in high-risk cohorts (kennel/veterinary/lab workers, owners of seropositive dogs) to quantify true human burden.
4. **Import-screening policy pilot** with mandatory pre-import *B. canis* testing, coupled to WGS/MLVA typing to map introduction routes (extending PMID: 33965789).
5. **Vaccine research** exploiting rough-LPS/T4SS/BvrR-BvrS biology to design attenuated or subunit candidates with a defined immune correlate.
6. **Mechanistic dissection** of tissue tropism (placenta, disc, uvea) using canine primary cells/organoids to explain why bacteremia seeds these specific sites.

Report compiled from 12 confirmed findings across 28 reviewed papers. Evidence types: veterinary clinical, human clinical/seroepidemiological, in vitro macrophage models, and computational genomics.
